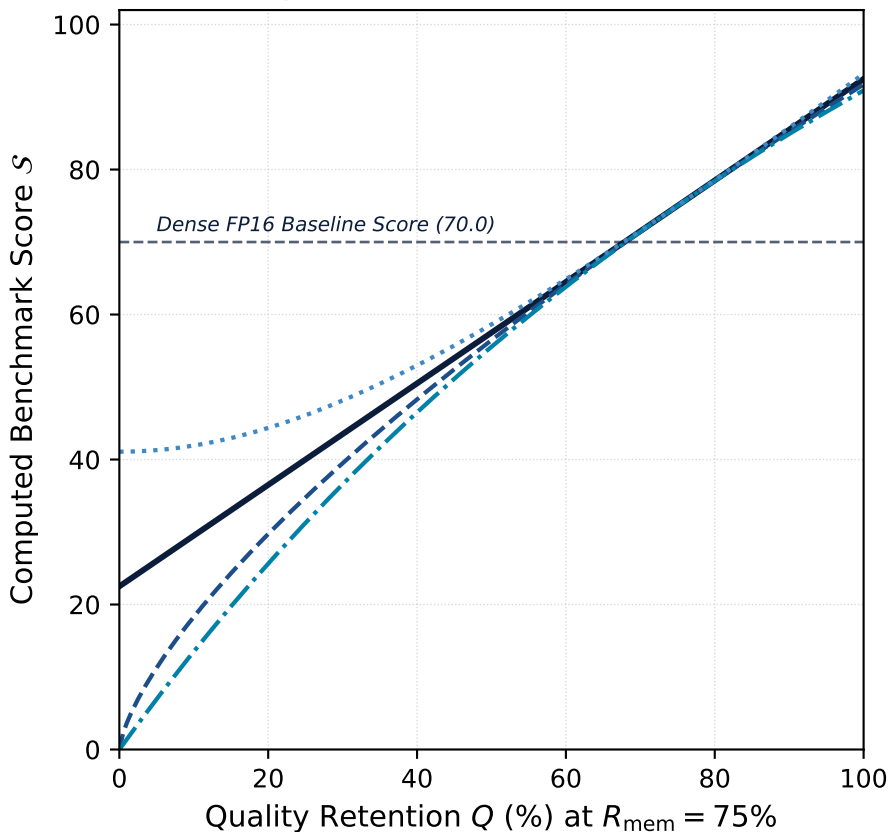


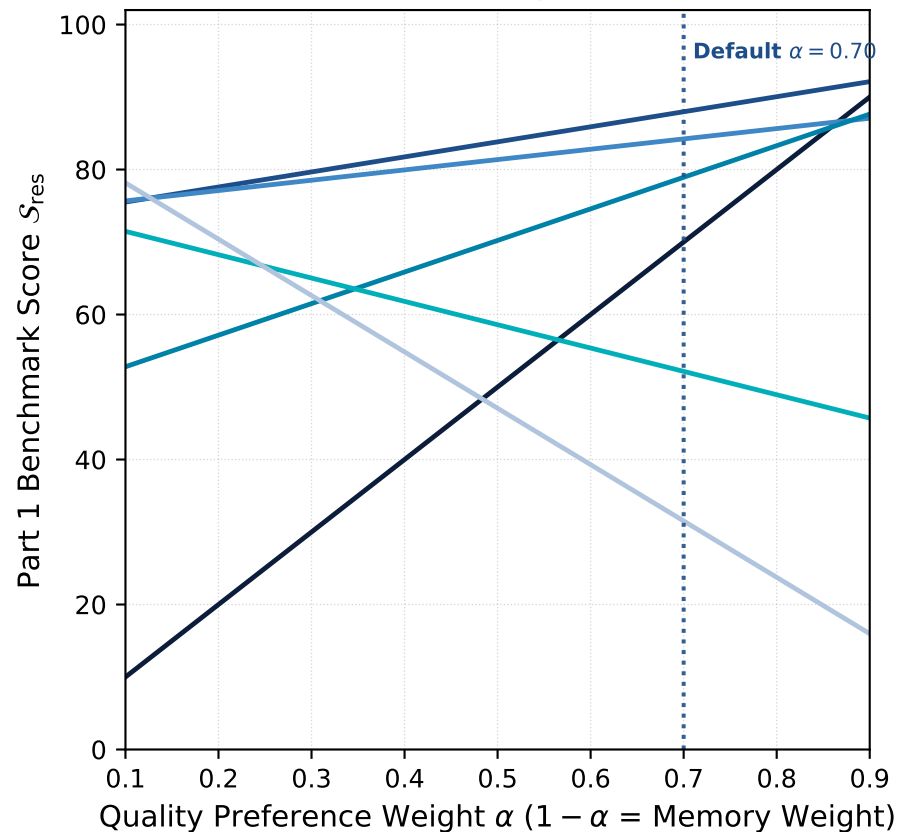
Mathematical Behavior and Sensitivity Analysis of the CRBench Linear Utility Formulation

(a) Utility Formulations at Fixed $R_{\text{mem}} = 75\%$



- Linear Additive $S = \alpha Q + (1 - \alpha)R$ [CRBench]
- Cobb-Douglas $S = Q^\alpha R^{1-\alpha}$
- Harmonic Mean
- Power Mean ($p = 2$)

(b) Method Score Sensitivity across $\alpha \in [0.10, 0.90]$



- Dense Baseline (FP16)
- DKV (Mid Preset)
- Low-Rank KV (SVD)
- KV Quant INT8
- SnapKV (Eviction)
- KV Quant INT2 (Collapsing)